

**SIMATS ENGINEERING**  
**Department of Computer Science & Engineering**  
**ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK**

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|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------|
| <b>Course Code:</b>   | <b>CSA1204</b>                                                                                                                                                                                                                                            | <b>Course Title:</b> | <b>Computer Architecture</b>                                      |
| <b>CO Assessed:</b>   | <b>CO4</b>                                                                                                                                                                                                                                                | <b>Assessment:</b>   | <b>ASSESSMENT<br/>TOOL1-INDUSTRY<br/>PROBLEM SOLVING<br/>TASK</b> |
| <b>Weightage:</b>     | <b>30%</b>                                                                                                                                                                                                                                                | <b>Total Marks:</b>  | <b>25</b>                                                         |
| <b>CO4 Statement:</b> | <i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i> |                      |                                                                   |
| <b>Student Name:</b>  | KAMALESH K                                                                                                                                                                                                                                                | <b>Reg.No.:</b>      | 192572164                                                         |
| <b>Department:</b>    | B.E.CSE(AI)                                                                                                                                                                                                                                               | <b>Date:</b>         | 22\08\2026                                                        |

**ANNEXURE A**  
**INDUSTRY PROBLEM SOLVING TASK**

- Smartphone Performance Optimization**  
A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.
- Cloud Server Memory Bottleneck**  
A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.
- Autonomous Vehicle Data Processing**  
An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.
- AI Inference Edge Device**  
An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.
- High-Performance Gaming Console**  
A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

**Code of Conduct:**

*I / Kavinadhithya R S [192511365] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*

**MARKS ALLOCATION**

|    | Understanding of Industry Problem (20%) | Application of Engineering Concepts (25%) | Solution Design (25%) | Use of Tools (15%) | Analysis (10%) |
|----|-----------------------------------------|-------------------------------------------|-----------------------|--------------------|----------------|
| Q1 |                                         |                                           |                       |                    |                |
| Q2 |                                         |                                           |                       |                    |                |
| Q3 |                                         |                                           |                       |                    |                |
| Q4 |                                         |                                           |                       |                    |                |
| Q5 |                                         |                                           |                       |                    |                |

**RUBRICS FOR CALCULATION:**

| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
|-------------------------------------|-----------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Understanding of Industry Problem   | 20%       | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25%       | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25%       | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15%       | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10%       | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |



# Q1. Smartphone Performance Optimization

## Problem Analysis

A smartphone experiences lag while switching between applications. The main reason can be slow movement of frequently used instructions and data between the CPU and main memory. The memory hierarchy should therefore be designed to keep frequently accessed data closer to the CPU.

Role of L1, L2, L3 Cache and DRAM

| Memory   | Speed            | Size         | Main Purpose                                  |
|----------|------------------|--------------|-----------------------------------------------|
| L1 Cache | Very high        | Very small   | Stores immediately required data/instructions |
| L2 Cache | High             | Small/medium | Stores recently used data                     |
| L3 Cache | Medium           | Larger       | Shared cache for multiple CPU cores           |
| DRAM     | Lower than cache | Large        | Stores applications and operating data        |

## Proposed Memory Hierarchy

CPU → L1 Cache → L2 Cache → L3 Cache → DRAM → Storage

- L1 cache: Store the most frequently accessed instructions and data.
- L2 cache: Store recently used application data.
- L3 cache: Provide a larger shared cache for multiple CPU cores.
- DRAM: Store running applications and larger datasets.

## Recommended Solution

The smartphone should use a larger and efficient L2/L3 cache, high-speed LPDDR DRAM, and an effective cache replacement/prefetching mechanism. Frequently used application data should remain in cache as much as possible.

## Expected Result

This design reduces memory access latency, improves CPU utilization and makes app switching faster and smoother.

## Q2. Cloud Server Memory Bottleneck

### Problem Analysis

A cloud server experiences high latency during database queries. Database operations repeatedly access frequently used data. If the required data is not available in cache, the processor has to access DRAM or even secondary storage through virtual memory paging, increasing latency.

### Cache Hierarchy vs Virtual Memory Paging

| Feature      | Cache Hierarchy               | Virtual Memory Paging       |
|--------------|-------------------------------|-----------------------------|
| Purpose      | Reduce CPU memory access time | Extend available memory     |
| Speed        | Very fast                     | Slower                      |
| Technology   | SRAM                          | DRAM + SSD/HDD              |
| Unit         | Cache line                    | Page                        |
| Main benefit | Reduces latency               | Allows larger programs/data |
| Suitable for | Frequently accessed data      | Data that cannot fit in RAM |

### Recommended Memory Structure

CPU → L1 → L2 → L3 → DRAM → SSD

- Keep frequently accessed database indexes in cache.
- Use sufficient DRAM to reduce page faults.
- Use SSD instead of HDD for paging when paging is unavoidable.
- Increase L3 cache where workload benefits from it.
- Optimize database queries and indexes.

### Recommended Solution

The cloud provider should prioritize cache optimization and sufficient DRAM rather than depending heavily on virtual-memory paging. Paging should be minimized because accessing secondary storage is much slower.

### Expected Result

The proposed system reduces cache misses and page faults, resulting in lower database-query latency and higher server throughput.

### Q3. Autonomous Vehicle Data Processing

#### Problem Analysis

An autonomous vehicle continuously receives data from cameras, radar, LiDAR and other sensors. This data must be processed in real time. Therefore, the memory system requires high bandwidth, low latency and reliable operation.

Comparison of SRAM, DRAM and MRAM

| Feature                                                       | SRAM           | DRAM             | MRAM        |
|---------------------------------------------------------------|----------------|------------------|-------------|
| Speed                                                         | Very high      | High             | High        |
| Latency                                                       | Very low       | Higher than SRAM | Low         |
| Volatile                                                      | Yes            | Yes              | No          |
| Density                                                       | Low            | High             | Medium      |
| Power                                                         | Higher per bit | Moderate         | Low standby |
| Main Use CPU cache    Main memory    Fast non-volatile memory |                |                  |             |

#### Proposed Memory Hierarchy

CPU/AI Accelerator → SRAM Cache → DRAM → MRAM → Storage  
Design

1. SRAM

- Use for L1/L2 cache.
- Store frequently processed sensor data.
- Provides extremely low latency.

2. DRAM

- Use as the main working memory.

- Store large camera, LiDAR and radar datasets.
- Provides higher capacity.

### 3. MRAM

- Use for non-volatile storage of important configuration data and frequently required information.
- It retains data without continuous power.

### Recommended Solution

A combination of SRAM + high-bandwidth DRAM + MRAM provides a good balance between latency, bandwidth and non-volatility.

### Expected Result

This hierarchy allows sensor data to be processed quickly, reducing processing delays and supporting real-time autonomous driving decisions.

## Q4. AI Inference Edge Device

### Problem Analysis

An edge AI device needs to load AI models quickly while consuming less power. A suitable memory hierarchy should combine fast volatile memory with high-density and non-volatile memory.

Comparison

| Memory     | Speed       | Power    | Volatility   | Main Use                  |
|------------|-------------|----------|--------------|---------------------------|
| NAND Flash | Medium/slow | Low      | Non-volatile | Model storage             |
| DRAM       | High        | Moderate | Volatile     | Running model             |
| RRAM       | High        | Low      | Non-volatile | Fast model/weight storage |

### Proposed Hierarchy

AI Processor → SRAM/Cache → DRAM → RRAM → NAND Flash

Design

NAND Flash

- Stores the complete AI model.
- Provides high storage capacity.

RRAM

- Stores frequently used model weights.
- Provides non-volatile and low-power access.

DRAM

- Loads active portions of the AI model.
- Provides fast working memory.

Recommended Solution

The edge device should use NAND Flash for bulk storage, RRAM for frequently accessed model data, and DRAM for active inference operations.

Expected Result

This structure reduces model-loading time and power consumption while maintaining fast AI inference performance.

Q5. High-Performance Gaming Console

Problem Analysis

A gaming console experiences frame drops while loading large game scenes. Large textures, models, audio files and game assets must be transferred rapidly into memory. Insufficient memory bandwidth or cache misses can cause delays.

L3 Cache vs DRAM

| Feature  | L3 Cache  | DRAM             |
|----------|-----------|------------------|
| Speed    | Very high | Lower than cache |
| Capacity | Smaller   | Much larger      |



|                   |                          |                     |
|-------------------|--------------------------|---------------------|
| Feature           | L3 Cache                 | DRAM                |
| Technology        | SRAM                     | DRAM                |
| Latency           | Low                      | Higher              |
| Purpose           | Frequently used CPU data | Large game datasets |
| Cost per bit High |                          | Lower               |

### Proposed Memory Hierarchy

CPU/GPU → L1 → L2 → L3 → High-Bandwidth DRAM → SSD

Improvements

1. Increase the effective L3 cache capacity.
2. Use high-bandwidth DRAM.
3. Improve cache prefetching.
4. Keep frequently accessed game assets in cache.
5. Use fast SSD storage for large game files.
6. Transfer game assets into DRAM before they are required.
7. Optimize memory access between CPU, GPU and DRAM.

### Recommended Solution

L3 cache should handle frequently reused game data, while high-bandwidth DRAM should store large textures, maps and other game assets. Fast SSD storage can provide rapid loading of data into DRAM.

### Expected Result

The improved hierarchy reduces memory stalls and improves data-transfer speed, resulting in smoother frame rates and fewer frame drops.